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**Note: Here All Python Assignment is Considered.**

**Assignments (No Loops), For Loop and While Loop, Using String Methods, Using List Methods**

# Python Conditional Statements – Assignments (No Loops)

**1. Write a program to check whether a number is positive or negative.**
**Code**

```
n = int(input('Enter a number: '))
if n >= 0:
    print('Positive' if n > 0 else 'Zero')
else:
    print('Negative')
```

**Output**

```
Enter a number: 12
Positive
```

**2. Write a program to check whether a number is even or odd.**
**Code**

```
n = int(input('Enter a number: '))
print('Even' if n % 2 == 0 else 'Odd')
```

**Output**

```
Enter a number: 8
Even
```

**3. Write a program to check whether a person is eligible to vote or not.**
**Code**

```
age = int(input('Enter age: '))
print('Eligible to vote' if age >= 18 else 'Not eligible to vote')
```

**Output**

```
Enter age: 20
Eligible to vote
```

**4. Write a program to check whether a number is greater than 100 or not.**
**Code**

```
n = int(input('Enter a number: '))
print('Greater than 100' if n > 100 else 'Not greater than 100')
```

**Output**

```
Enter a number: 125
Greater than 100
```

**5. Write a program to check whether a given year is a leap year or not.**
**Code**

```
year = int(input('Enter year: '))
if year % 400 == 0 or (year % 4 == 0 and year % 100 != 0): print('Leap year')
else: print('Not a leap year')
```

**Output**

```
Enter year: 2024
Leap year
```

**6. Write a program to compare two numbers and print the greater one.**
**Code**

```
a = int(input('Enter first number: '))
b = int(input('Enter second number: '))
if a > b: print(a, 'is greater')
elif b > a: print(b, 'is greater')
else: print('Both are equal')
```

**Output**

```
Enter first number: 12
Enter second number: 9
12 is greater
```

**7. Write a program to check whether a character is a vowel or consonant.**
**Code**

```
ch = input('Enter a character: ').lower()
print('Vowel' if ch in 'aeiou' else 'Consonant')
```

**Output**

```
Enter a character: a
Vowel
```

**8. Write a program to check whether a number is divisible by 5 or not.**
**Code**

```
n = int(input('Enter a number: '))
print('Divisible by 5' if n % 5 == 0 else 'Not divisible by 5')
```

**Output**

```
Enter a number: 25
Divisible by 5
```

**9. Write a program to check whether a student has passed or failed (pass marks = 40).**
**Code**

```
marks = float(input('Enter marks: '))
print('Pass' if marks >= 40 else 'Fail')
```

**Output**

```
Enter marks: 75
Pass
```

**10. Write a program to check whether a number is a single-digit number or not.**
**Code**

```
n = int(input('Enter a number: '))
print('Single-digit number' if -9 <= n <= 9 else 'Not a single-digit number')
```

**Output**

```
Enter a number: 7
Single-digit number
```

**11. Write a program to check whether a temperature is hot (>30) or cold.**
**Code**

```
temp = float(input('Enter temperature: '))
print('Hot' if temp > 30 else 'Cold or moderate')
```

**Output**

```
Enter temperature: 35
Hot
```

**12. Write a program to check whether a person is a child or adult (age < 18).**
**Code**

```
age = int(input('Enter age: '))
print('Child' if age < 18 else 'Adult')
```

**Output**

```
Enter age: 17
Child
```

**13. Write a program to check whether a number is zero or non-zero.**
**Code**

```
n = int(input('Enter a number: '))
print('Zero' if n == 0 else 'Non-zero')
```

**Output**

```
Enter a number: 5
Non-zero
```

**14. Write a program to check whether a given character is uppercase or lowercase.**
**Code**

```
ch = input('Enter a character: ')
if ch.isupper(): print('Uppercase')
elif ch.islower(): print('Lowercase')
else: print('Not an alphabetic case')
```

**Output**

```
Enter a character: K
Uppercase
```

**15. Write a program to check whether a number is divisible by both 3 and 7 or not.**
**Code**

```
n = int(input('Enter a number: '))
print('Divisible by both 3 and 7' if n % 3 == 0 and n % 7 == 0 else 'Not divisible by both 3 and 7')
```

**Output**

```
Enter a number: 21
Divisible by both 3 and 7
```

**16. Write a program to find the greatest of three numbers.**
**Code**

```
a, b, c = map(int, input('Enter three numbers: ').split())
print('Greatest:', max(a, b, c))
```

**Output**

```
Enter three numbers: 12 25 18
Greatest: 25
```

**17. Write a program to check grade based on marks (A, B, C, Fail).**
**Code**

```
marks = float(input('Enter marks: '))
if marks >= 80: grade = 'A'
elif marks >= 60: grade = 'B'
elif marks >= 40: grade = 'C'
else: grade = 'Fail'
print('Grade:', grade)
```

**Output**

```
Enter marks: 85
Grade: A
```

**18. Write a program to check whether a number is positive, negative, or zero.**
**Code**

```
n = int(input('Enter a number: '))
if n > 0: print('Positive')
elif n < 0: print('Negative')
else: print('Zero')
```

**Output**

```
Enter a number: -4
Negative
```

**19. Write a program to print the day name based on day number (1–7).**
**Code**

```
day = int(input('Enter day number (1-7): '))
days = {1:'Monday', 2:'Tuesday', 3:'Wednesday', 4:'Thursday', 5:'Friday', 6:'Saturday', 7:'Sunday'}
print(days.get(day, 'Invalid day number'))
```

**Output**

```
Enter day number (1-7): 3
Wednesday
```

**20. Write a program to check ticket price based on age.**
**Code**

```
age = int(input('Enter age: '))
price = 0 if age < 5 else 100 if age < 60 else 60
print('Ticket price: ₹', price)
```

**Output**

```
Enter age: 25
Ticket price: ₹ 100
```

**21. Write a program to find the smallest of three numbers.**
**Code**

```
a, b, c = map(int, input('Enter three numbers: ').split())
print('Smallest:', min(a, b, c))
```

**Output**

```
Enter three numbers: 12 25 18
Smallest: 12
```

**22. Write a program to check salary bonus based on years of service.**
**Code**

```
salary = float(input('Enter salary: '))
years = int(input('Years of service: '))
bonus = salary * 0.10 if years >= 5 else 0
print('Bonus: ₹', bonus)
```

**Output**

```
Enter salary: 30000
Years of service: 6
Bonus: ₹ 3000.0
```

**23. Write a program to calculate electricity bill based on units consumed.**
**Code**

```
units = float(input('Enter units consumed: '))
if units <= 100: bill = units * 1.5
elif units <= 200: bill = 100*1.5 + (units-100)*2.5
else: bill = 100*1.5 + 100*2.5 + (units-200)*4
print('Electricity bill: ₹', bill)
```

**Output**

```
Enter units consumed: 150
Electricity bill: ₹ 275.0
```

**24. Write a program to check exam results (Distinction, First Class, Pass, Fail).**
**Code**

```
marks = float(input('Enter percentage: '))
if marks >= 75: result = 'Distinction'
elif marks >= 60: result = 'First Class'
elif marks >= 40: result = 'Pass'
else: result = 'Fail'
print(result)
```

**Output**

```
Enter percentage: 78
Distinction
```

**25. Write a program to check whether a triangle is equilateral, isosceles, or scalene.**
**Code**

```
a, b, c = map(float, input('Enter three sides: ').split())
if a == b == c: kind = 'Equilateral'
elif a == b or b == c or a == c: kind = 'Isosceles'
else: kind = 'Scalene'
print(kind, 'triangle')
```

**Output**

```
Enter three sides: 5 5 5
Equilateral triangle
```

**26. Write a program to check month names based on month number.**
**Code**

```
month = int(input('Enter month number: '))
months =
{1:'January',2:'February',3:'March',4:'April',5:'May',6:'June',7:'July',8:'August',9:'September',10:'Oct
ober',11:'November',12:'December'}
print(months.get(month, 'Invalid month'))
```

**Output**

```
Enter month number: 10
October
```

**27. Write a program to check login status based on username and password.**
**Code**

```
username = input('Username: ')
password = input('Password: ')
if username == 'Sumit' and password == 'python123': print('Login successful')
else: print('Invalid username or password')
```

**Output**

```
Username: Sumit
Password: python123
Login successful
```

**28. Write a program to calculate discounts based on total shopping amount.**
**Code**

```
amount = float(input('Shopping amount: ₹'))
discount = 0.10 if amount >= 5000 else 0.05 if amount >= 2000 else 0
print('Discount: ₹', amount * discount)
print('Payable: ₹', amount * (1-discount))
```

**Output**

```
Shopping amount: ₹3000
Discount: ₹ 150.0
Payable: ₹ 2850.0
```

**29. Write a program to check blood donation eligibility based on age and weight.**
**Code**

```
age = int(input('Enter age: '))
weight = float(input('Enter weight in kg: '))
print('Eligible' if age >= 18 and weight >= 50 else 'Not eligible')
```

**Output**

```
Enter age: 22
Enter weight in kg: 60
Eligible
```

**30. Write a program to check the type of number (single, double, or three digit).**
**Code**

```
n = abs(int(input('Enter an integer: ')))
if n <= 9: kind = 'Single-digit'
elif n <= 99: kind = 'Double-digit'
elif n <= 999: kind = 'Three-digit'
else: kind = 'More than three digits'
print(kind, 'number')
```

**Output**

```
Enter an integer: 45
Double-digit number
```

**31. Write a program to check whether a number is positive and even using nested if.**
**Code**

```
n = int(input('Enter a number: '))
if n > 0:
   if n % 2 == 0: print('Positive and even')
   else: print('Positive but odd')
else: print('Not positive')
```

**Output**

```
Enter a number: 8
Positive and even
```

**32. Write a program to check whether a student is eligible for an exam (attendance + marks).**
**Code**

```
attendance = float(input('Attendance percentage: '))
marks = float(input('Marks: '))
print('Eligible for exam' if attendance >= 75 and marks >= 40 else 'Not eligible for exam')
```

**Output**

```
Attendance percentage: 85
Marks: 65
Eligible for exam
```

**33. Write a program to check whether a person can apply for a loan.**
**Code**

```
salary = float(input('Monthly salary: ₹'))
age = int(input('Age: '))
print('Can apply for loan' if salary >= 15000 and age >= 21 else 'Cannot apply for loan')
```

**Output**

```
Monthly salary: ₹25000
Age: 25
Can apply for loan
```

**34. Write a program to check whether a year is a leap year and century year.**
**Code**

```
year = int(input('Enter year: '))
leap = year % 400 == 0 or (year % 4 == 0 and year % 100 != 0)
print('Leap year' if leap else 'Not a leap year')
print('Century year' if year % 100 == 0 else 'Not a century year')
```

**Output**

```
Enter year: 2000
Leap year
Century year
```

**35. Write a program to check whether a number is divisible by 3 and 5.**
**Code**

```
n = int(input('Enter a number: '))
print('Divisible by both 3 and 5' if n % 3 == 0 and n % 5 == 0 else 'Not divisible by both 3 and 5')
```

**Output**

```
Enter a number: 30
Divisible by both 3 and 5
```

**36. Write a program to check whether a person is eligible for a driving license.**
**Code**

```
age = int(input('Enter age: '))
print('Eligible for driving license' if age >= 18 else 'Not eligible for driving license')
```

**Output**

```
Enter age: 20
Eligible for driving license
```

**37. Write a program to check whether a number lies between 10 and 50.**
**Code**

```
n = int(input('Enter a number: '))
print('Within range' if 10 <= n <= 50 else 'Outside range')
```

**Output**

```
Enter a number: 25
Within range
```

**38. Write a program to check whether a student passed all subjects.**
**Code**

```
marks = list(map(float, input('Enter marks for 3 subjects: ').split()))
print('Passed all subjects' if all(m >= 40 for m in marks) else 'Failed in at least one subject')
```

**Output**

```
Enter marks for 3 subjects: 70 65 80
Passed all subjects
```

**39. Write a program to check whether a number is greater than 0 and less than 100.**
**Code**

```
n = int(input('Enter a number: '))
print('Greater than 0 and less than 100' if 0 < n < 100 else 'Outside the range')
```

**Output**

```
Enter a number: 50
Greater than 0 and less than 100
```

**40. Write a program to check whether a password is valid using nested if.**
**Code**

```
password = input('Enter password: ')
if len(password) >= 8:
    if any(c.isdigit() for c in password): print('Valid password')
    else: print('Password must contain a digit')
else: print('Password must be at least 8 characters')
```

**Output**

```
Enter password: Sumit123
Valid password
```

**41. Write a program to print weekday name using match case.**
**Code**

```
day = int(input('Enter weekday number (1-7): '))
match day:
    case 1: print('Monday')
    case 2: print('Tuesday')
    case 3: print('Wednesday')
    case 4: print('Thursday')
    case 5: print('Friday')
    case 6: print('Saturday')
    case 7: print('Sunday')
    case _: print('Invalid day')
```

**Output**

```
Enter weekday number (1-7): 1
Monday
```

**42. Write a program to perform calculator operations using match case.**
**Code**

```
a = float(input('First number: ')); op = input('Operator (+, -, *, /): '); b = float(input('Second
number: '))
match op:
    case '+': print(a+b)
    case '-': print(a-b)
    case '*': print(a*b)
    case '/' if b != 0: print(a/b)
    case '/': print('Cannot divide by zero')
    case _: print('Invalid operator')
```

**Output**

```
First number: 8
Operator (+, -, *, /): *
```

```
Second number: 3
24.0
```

**43. Write a program to print month days using match case.**
**Code**

```
month = int(input('Enter month number: '))
match month:
    case 2: print('28 or 29 days')
    case 4 | 6 | 9 | 11: print('30 days')
    case 1 | 3 | 5 | 7 | 8 | 10 | 12: print('31 days')
    case _: print('Invalid month')
```

**Output**

```
Enter month number: 2
28 or 29 days
```

**44. Write a program to check traffic signal actions using match case.**
**Code**

```
signal = input('Enter traffic signal color: ').lower()
match signal:
    case 'red': print('Stop')
    case 'yellow': print('Get ready')
    case 'green': print('Go')
    case _: print('Invalid signal')
```

**Output**

```
Enter traffic signal color: red
Stop
```

**45. Write a program to print menu-based food selection using match case.**
**Code**

```
choice = int(input('1. Poha  2. Pizza  3. Sandwich: '))
match choice:
    case 1: print('You selected Poha')
    case 2: print('You selected Pizza')
    case 3: print('You selected Sandwich')
    case _: print('Invalid choice')
```

**Output**

```
1. Poha  2. Pizza  3. Sandwich: 1
You selected Poha
```

**46. Write a program to print grade description using match case.**
**Code**

```
grade = input('Enter grade (A/B/C/F): ').upper()
match grade:
```

```
    case 'A': print('Excellent')
    case 'B': print('Very good')
    case 'C': print('Good')
    case 'F': print('Fail')
    case _: print('Invalid grade')
```

**Output**

```
Enter grade (A/B/C/F): A
Excellent
```

**47. Write a program to check user role (admin, user, guest) using match case.**
**Code**

```
role = input('Enter role: ').lower()
match role:
    case 'admin': print('Full access')
    case 'user': print('Standard access')
    case 'guest': print('Limited access')
    case _: print('Unknown role')
```

**Output**

```
Enter role: admin
Full access
```

**48. Write a program to print language choice using match case.**
**Code**

```
choice = input('Choose language (English/Marathi/Hindi): ').lower()
match choice:
    case 'english': print('Language selected: English')
    case 'marathi': print('Language selected: Marathi')
    case 'hindi': print('Language selected: Hindi')
    case _: print('Invalid choice')
```

**Output**

```
Choose language (English/Marathi/Hindi): Marathi
Language selected: Marathi
```

**49. Write a program to print result status using match case.**
**Code**

```
marks = float(input('Enter marks: '))
match True:
    case _ if marks >= 40: print('Pass')
    case _: print('Fail')
```

**Output**

```
Enter marks: 72
Pass
```

**50. Write a program to build a simple ATM menu using a match case.**
**Code**

```
balance = 5000
choice = int(input('1. Balance  2. Deposit  3. Withdraw: '))
match choice:
    case 1: print('Balance: ₹', balance)
    case 2: print('Deposit feature selected')
    case 3: print('Withdraw feature selected')
    case _: print('Invalid choice')
```

**Output**

```
1. Balance  2. Deposit  3. Withdraw: 1
Balance: ₹ 5000
```

# Python Assignments: For Loop and While Loop

**1. Print numbers from 1 to 10 using for loop.**
**Code**

```
for i in range(1, 11): print(i, end=' ')
```

**Output**

```
1 2 3 4 5 6 7 8 9 10
```

**2. Print numbers from 10 to 1 using for loop.**
**Code**

```
for i in range(10, 0, -1): print(i, end=' ')
```

**Output**

```
10 9 8 7 6 5 4 3 2 1
```

**3. Print all even numbers between 1 and 50.**
**Code**

```
for i in range(2, 51, 2): print(i, end=' ')
```

**Output**

```
2 4 6 8 10 ... 50
```

**4. Print all odd numbers between 1 and 50.**
**Code**

```
for i in range(1, 51, 2): print(i, end=' ')
```

**Output**

```
1 3 5 7 9 ... 49
```

**5. Print multiplication table of a given number.**
**Code**

```
n = int(input('Enter number: '))
for i in range(1, 11): print(f'{n} x {i} = {n*i}')
```

**Output**

```
Enter number: 5
5 x 1 = 5
5 x 2 = 10
...
5 x 10 = 50
```

**6. Calculate sum of first 10 natural numbers.**
**Code**

```
print(sum(range(1, 11)))
```

**Output**

```
55
```

**7. Calculate factorial of a number using for loop.**
**Code**

```
n = int(input('Enter number: '))
fact = 1
for i in range(1, n+1): fact *= i
print(fact)
```

**Output**

```
Enter number: 5
120
```

**8. Print all characters of a string.**
**Code**

```
s = input('Enter string: ')
for ch in s: print(ch)
```

**Output**

```
Enter string: Sumit
K
u
n
d
a
n
```

**9. Print square of numbers from 1 to 10.**
**Code**

```
for i in range(1, 11): print(i, end=' ')
```

**Output**

```
1 2 3 4 5 6 7 8 9 10
```

**10. Print cube of numbers from 1 to 10.**
**Code**

```
for i in range(1, 11): print(i, end=' ')
```

**Output**

```
1 2 3 4 5 6 7 8 9 10
```

**11. Count number of digits in a number.**
**Code**

```
n = abs(int(input('Enter number: ')))
print(len(str(n)))
```

**Output**

```
Enter number: 12345
5
```

**12. Reverse a number using while loop.**
**Code**

```
n = int(input('Enter number: ')); rev = 0
while n > 0:
    rev = rev*10 + n%10
    n //= 10
print(rev)
```

**Output**

```
Enter number: 1234
4321
```

**13. Check whether a number is palindrome.**
**Code**

```
n = input('Enter number: ')
print('Palindrome' if n == n[::-1] else 'Not palindrome')
```

**Output**

```
Enter number: 121
Palindrome
```

**14. Print Fibonacci series up to n terms.**
**Code**

```
n = int(input('Terms: ')); a, b = 0, 1
for _ in range(n): print(a, end=' '); a, b = b, a+b
```

**Output**

```
Terms: 7
0 1 1 2 3 5 8
```

**15. Find sum of digits of a number.**
**Code**

```
n = abs(int(input('Enter number: '))); total = 0
while n: total += n%10; n //= 10
print(total)
```

**Output**

```
Enter number: 1234
10
```

**16. Print numbers divisible by 5 between 1 and 100.**
**Code**

```
for i in range(5, 101, 5): print(i, end=' ')
```

**Output**

```
5 10 15 ... 100
```

**17. Find largest digit in a number.**
**Code**

```
n = input('Enter number: ')
print(max(n))
```

**Output**

```
Enter number: 5832
8
```

**18. Print prime numbers between 1 and 50.**
**Code**

```
for n in range(2, 51):
   if all(n % i for i in range(2, int(n**0.5)+1)): print(n, end=' ')
```

**Output**

```
2 3 5 7 11 13 ... 47
```

**19. Check whether a number is prime.**
**Code**

```
n = int(input('Enter number: '))
print('Prime' if n > 1 and all(n%i for i in range(2, int(n**0.5)+1)) else 'Not prime')
```

**Output**

```
Enter number: 7
Prime
```

**20. Print pattern of stars (triangle).**
**Code**

```
for i in range(1, 5): print('*' * i)
```

**Output**

```
*
**
***
****
```

**21. Print pattern of numbers.**
**Code**

```
for i in range(1, 5): print(''.join(str(j) for j in range(1, i+1)))
```

**Output**

```
1
12
123
1234
```

**22. Find sum of even digits in a number.**
**Code**

```
n = input('Enter number: ')
print(sum(int(d) for d in n if int(d)%2==0))
```

**Output**

```
Enter number: 123456
12
```

**23. Find sum of odd digits in a number.**
**Code**

```
n = input('Enter number: ')
print(sum(int(d) for d in n if int(d)%2==1))
```

**Output**

```
Enter number: 123456
9
```

**24. Count vowels in a string.**
**Code**

```
s = input('Enter string: ').lower()
print(sum(c in 'aeiou' for c in s))
```

**Output**

```
Enter string: Sumit
2
```

**25. Count consonants in a string.**
**Code**

```
s = input('Enter string: ').lower()
print(sum(c.isalpha() and c not in 'aeiou' for c in s))
```

**Output**

```
Enter string: Sumit
4
```

**26. Print ASCII values of characters in a string.**
**Code**

```
s = input('Enter string: ')
for c in s: print(c, ord(c))
```

**Output**

```
Enter string: K
K 75
```

**27. Print all factors of a number.**
**Code**

```
n = int(input('Enter number: '))
for i in range(1, n+1):
    if n % i == 0: print(i, end=' ')
```

**Output**

```
Enter number: 12
1 2 3 4 6 12
```

**28. Check whether a number is Armstrong.**
**Code**

```
n = input('Enter number: ')
print('Armstrong' if sum(int(d)**len(n) for d in n) == int(n) else 'Not Armstrong')
```

**Output**

```
Enter number: 153
Armstrong
```

**29. Generate first n natural numbers using while loop.**
**Code**

```
n = int(input('Enter n: ')); i = 1
while i <= n: print(i, end=' '); i += 1
```

**Output**

```
Enter n: 5
1 2 3 4 5
```

**30. Generate first n even numbers.**
**Code**

```
n = int(input('Enter n: '))
for i in range(1, n+1): print(i*2, end=' ')
```

**Output**

```
Enter n: 5
2 4 6 8 10
```

**31. Generate first n odd numbers.**
**Code**

```
n = int(input('Enter n: '))
for i in range(n): print(2*i+1, end=' ')
```

**Output**

```
Enter n: 5
1 3 5 7 9
```

**32. Calculate power of a number without using pow().**
**Code**

```
base = int(input('Base: ')); exp = int(input('Exponent: ')); result = 1
for _ in range(exp): result *= base
print(result)
```

**Output**

```
Base: 2
Exponent: 5
32
```

**33. Print reverse of a string.**
**Code**

```
s = input('Enter string: ')
print(s[::-1])
```

**Output**

```
Enter string: Sumit
nadnuK
```

**34. Check whether a string is palindrome.**
**Code**

```
s = input('Enter string: ')
print('Palindrome' if s.lower()==s[::-1].lower() else 'Not palindrome')
```

**Output**

```
Enter string: level
Palindrome
```

**35. Print multiplication tables from 1 to 5.**
**Code**

```
for n in range(1, 6):
    for i in range(1, 11): print(f'{n}x{i}={n*i}', end='  ')
    print()
```

**Output**

```
1x1=1 ... 5x10=50
```

**36. Find sum of numbers entered by user until 0 is entered.**
**Code**

```
total = 0
while True:
    n = int(input('Enter number (0 to stop): '))
    if n == 0: break
    total += n
print('Sum:', total)
```

**Output**

```
Enter number (0 to stop): 5
Enter number (0 to stop): 7
Enter number (0 to stop): 0
Sum: 12
```

**37. Print numbers until user enters a negative number.**
**Code**

```
while True:
    n = int(input('Enter number: '))
    if n < 0: break
    print(n)
```

**Output**

```
Enter number: 3
3
Enter number: -1
```

**38. Find smallest digit in a number.**
**Code**

```
n = input('Enter number: ')
print(min(n))
```

**Output**

```
Enter number: 5832
2
```

**39. Count total words in a string.**
**Code**

```
s = input('Enter sentence: ')
print(len(s.split()))
```

**Output**

```
Enter sentence: Sumit studies Python
3
```

**40. Print numbers which are perfect squares between 1 and 100.**
**Code**

```
print([i*i for i in range(1, 11)])
```

**Output**

```
[1, 4, 9, 16, 25, 36, 49, 64, 81, 100]
```

**41. Print numbers which are perfect cubes between 1 and 100**
**Code**

```
print([i**3 for i in range(1, 5)])
```

**Output**

```
[1, 8, 27, 64]
```

**42. Calculate sum of squares of first n numbers.**
**Code**

```
n = int(input('Enter n: '))
print(sum(i*i for i in range(1, n+1)))
```

**Output**

```
Enter n: 3
14
```

**43. Calculate sum of cubes of first n numbers.**
**Code**

```
n = int(input('Enter n: '))
print(sum(i**3 for i in range(1, n+1)))
```

**Output**

```
Enter n: 3
36
```

**44. Print alternate characters of a string.**
**Code**

```
s = input('Enter string: ')
for ch in s: print(ch)
```

**Output**

```
Enter string: Sumit
K
u
n
d
a
n
```

**45. Print numbers skipping multiples of 3.**
**Code**

```
for i in range(1, 21):
    if i % 3 != 0: print(i, end=' ')
```

**Output**

```
1 2 4 5 7 8 ... 20
```

**46. Print first n multiples of a number.**
**Code**

```
n = int(input('Enter number: ')); count = int(input('How many multiples: '))
for i in range(1, count+1): print(n*i, end=' ')
```

**Output**

```
Enter number: 5
How many multiples: 4
5 10 15 20
```

**47. Find product of digits of a number.**
**Code**

```
n = input('Enter number: '); product = 1
for d in n: product *= int(d)
print(product)
```

**Output**

```
Enter number: 234
24
```

**48. Check whether number is strong number.**
**Code**

```
import math
n = int(input('Enter number: '))
print('Strong number' if sum(math.factorial(int(d)) for d in str(n)) == n else 'Not a strong number')
```

**Output**

```
Enter number: 145
Strong number
```

**49. Display menu-driven program using while loop.**
**Code**

```
while True:
    print('1. Greet  2. Show name  3. Exit')
    choice = input('Choose: ')
    if choice == '1': print('Hello, Sumit!')
    elif choice == '2': print('Sumit Prakash Mane')
    elif choice == '3': break
    else: print('Invalid choice')
```

**Output**

1. Greet  2. Show name  3. Exit
Choose: 2
Sumit Prakash Mane

# Python String Assignments (Using String Methods)

**1. Take a string input and convert it to uppercase.**
**Code**

```
s = input('Enter a string: ')
print(s.upper())
```

**Output**

```
Enter a string: Sumit
SUMIT
```

**2. Take a string input and convert it to lowercase.**
**Code**

```
s = input('Enter a string: ')
print(s.lower())
```

**Output**

```
Enter a string: SUMIT
Sumit
```

**3. Count the total number of characters in a string.**
**Code**

```
s = input('Enter a string: ')
print('Characters:', len(s))
```

**Output**

```
Enter a string: Sumit
Characters: 6
```

**4. Count the number of vowels in a string.**
**Code**

```
s = input('Enter a string: ')
print('Vowels:', sum(ch.lower() in 'aeiou' for ch in s))
```

**Output**

```
Enter a string: Sumit
Vowels: 2
```

**5. Count the number of consonants in a string.**
**Code**

```
s = input('Enter a string: ')
print('Consonants:', sum(ch.isalpha() and ch.lower() not in 'aeiou' for ch in s))
```

**Output**

```
Enter a string: Sumit
Consonants: 4
```

**6. Check whether a string starts with a given substring.**
**Code**

```
s = input('Enter a string: ')
prefix = input('Enter prefix: ')
print(s.startswith(prefix))
```

**Output**

```
Enter a string: Sumit
Enter prefix: Kun
True
```

**7. Check whether a string ends with a given substring.**
**Code**

```
s = input('Enter a string: ')
suffix = input('Enter suffix: ')
print(s.endswith(suffix))
```

**Output**

```
Enter a string: Sumit
Enter suffix: dan
True
```

**8. Remove leading and trailing spaces from a string.**
**Code**

```
s = input('Enter a string: ')
print(s.strip())
```

**Output**

```
Enter a string:   Sumit
Sumit
```

**9. Replace all spaces in a string with hyphens (-).**
**Code**

```
s = input('Enter a string: ')
print(s.replace(' ', '-'))
```

**Output**

```
Enter a string: Sumit Mane
Sumit-Mane
```

**10. Find the index of a given character in a string.**
**Code**

```
s = input('Enter a string: ')
ch = input('Enter character: ')
print(s.find(ch))
```

**Output**

```
Enter a string: Sumit
Enter character: a
4
```

**11. Count how many times a particular character appears in a string.**
**Code**

```
s = input('Enter a string: ')
ch = input('Enter character: ')
print(s.count(ch))
```

**Output**

```
Enter a string: Sumit
Enter character: n
2
```

**12. Split a sentence into words using split().**
**Code**

```
s = input('Enter a sentence: ')
print(s.split())
```

**Output**

```
Enter a sentence: Sumit Mane studies Python
['Sumit', 'Mane', 'studies', 'Python']
```

**13. Join a list of words into a single string.**
**Code**

```
words = ['Sumit', 'studies', 'Python']
print(' '.join(words))
```

**Output**

```
Sumit studies Python
```

**14. Check whether a string is alphanumeric.**
**Code**

```
s = input('Enter a string: ')
print(s.isalnum())
```

**Output**

```
Enter a string: Sumit123
True
```

**15. Check whether a string contains only alphabets.**
**Code**

```
s = input('Enter a string: ')
print(s.isalpha())
```

**Output**

```
Enter a string: Sumit
True
```

**16. Check whether a string contains only digits.**
**Code**

```
s = input('Enter a string: ')
print(s.isdigit())
```

**Output**

```
Enter a string: 12345
True
```

**17. Capitalize the first character of a string.**
**Code**

```
s = input('Enter a string: ')
print(s.capitalize())
```

**Output**

```
Enter a string: Sumit mane
Sumit mane
```

**18. Convert the first character of each word to uppercase.**
**Code**

```
s = input('Enter a string: ')
print(s.title())
```

**Output**

```
Enter a string: Sumit mane
Sumit Mane
```

**19. Swap uppercase characters to lowercase and vice versa.**
**Code**

```
s = input('Enter a string: ')
print(s.swapcase())
```

**Output**

```
Enter a string: Sumit Mane
Sumit mANE
```

**20. Remove all occurrences of a given character from a string.**
**Code**

```
s = input('Enter a string: ')
ch = input('Character to remove: ')
print(s.replace(ch, ''))
```

**Output**

```
Enter a string: Sumit
Character to remove: n
Kuda
```

**21. Find the length of a string without using len().**
**Code**

```
s = input('Enter a string: ')
count = 0
for ch in s: count += 1
print(count)
```

**Output**

```
Enter a string: Sumit
6
```

**22. Reverse a string using string methods.**
**Code**

```
s = input('Enter a string: ')
print(s[::-1])
```

**Output**

```
Enter a string: Sumit
nadnuK
```

**23. Check whether a string is a palindrome.**
**Code**

```
s = input('Enter a string: ')
print('Palindrome' if s.lower() == s[::-1].lower() else 'Not palindrome')
```

**Output**

```
Enter a string: level
Palindrome
```

**24. Extract only digits from a string.**
**Code**

```
s = input('Enter a string: ')
print(s.isdigit())
```

**Output**

```
Enter a string: 12345
True
```

**25. Extract only alphabets from a string.**
**Code**

```
s = input('Enter a string: ')
print(s.isalpha())
```

**Output**

```
Enter a string: Sumit
True
```

**26. Find the first and last occurrence of a character.**
**Code**

```
s = input('Enter a string: ')
ch = input('Enter character: ')
print('First:', s.find(ch), 'Last:', s.rfind(ch))
```

**Output**

```
Enter a string: banana
Enter character: a
First: 1 Last: 5
```

**27. Check whether a word exists in a sentence.**
**Code**

```
s = input('Enter a sentence: ')
word = input('Enter word: ')
print(word in s.split())
```

**Output**

```
Enter a sentence: Sumit studies Python
Enter word: Python
True
```

**28. Count the number of words in a sentence.**
**Code**

```
s = input('Enter a sentence: ')
print('Words:', len(s.split()))
```

**Output**

```
Enter a sentence: Sumit studies Python
Words: 3
```

**29. Replace a word with another word in a sentence.**
**Code**

```
s = input('Enter a sentence: ')
print(s.replace('Python', 'coding'))
```

**Output**

```
Sumit likes Python
Sumit likes coding
```

**30. Check whether a string is in title case.**
**Code**

```
s = input('Enter a string: ')
print(s.title())
```

**Output**

```
Enter a string: Sumit mane
Sumit Mane
```

**31. Remove punctuation marks from a string.**
**Code**

```
import string
s = input('Enter a string: ')
print(''.join(ch for ch in s if ch not in string.punctuation))
```

**Output**

```
Enter a string: Sumit, Mane!
Sumit Mane
```

**32. Check whether all characters in a string are lowercase.**
**Code**

```
s = input('Enter a string: ')
print(s.islower())
```

**Output**

```
Enter a string: Sumit
True
```

**33. Check whether all characters in a string are uppercase.**
**Code**

```
s = input('Enter a string: ')
print(s.isupper())
```

**Output**

```
Enter a string: SUMIT
True
```

**34. Find the ASCII value of each character in a string.**
**Code**

```
s = input('Enter a string: ')
for ch in s: print(ch, ord(ch))
```

**Output**

```
Enter a string: K
K 75
```

**35. Convert a string into a list of characters.**
**Code**

```
s = input('Enter a string: ')
print(list(s))
```

**Output**

```
Enter a string: Sumit
['K', 'u', 'n', 'd', 'a', 'n']
```

**36. Remove duplicate characters from a string.**
**Code**

```
s = input('Enter a string: ')
result = ''.join(dict.fromkeys(s))
print(result)
```

**Output**

```
Enter a string: banana
ban
```

**37. Find the longest word in a sentence.**
**Code**

```
s = input('Enter a sentence: ')
print(max(s.split(), key=len))
```

**Output**

```
Enter a sentence: Sumit studies Python
studies
```

**38. Find the shortest word in a sentence.**
**Code**

```
s = input('Enter a sentence: ')
print(min(s.split(), key=len))
```

**Output**

```
Enter a sentence: Sumit is smart
is
```

**39. Check whether a string is a valid identifier.**
**Code**

```
s = input('Enter an identifier: ')
print(s.isidentifier())
```

**Output**

```
Enter an identifier: Sumit_name
True
```

**40. Count uppercase and lowercase letters separately.**
**Code**

```
s = input('Enter a string: ')
print(s.islower())
```

**Output**

```
Enter a string: Sumit
True
```

# Python List Assignments (Using List Methods)

**1. Create an empty list and add 5 integers using append().**
**Code**

```
items = []
for n in [10, 20, 30, 40, 50]: items.append(n)
print(items)
```

**Output**

```
[10, 20, 30, 40, 50]
```

**2. Create a list of strings and add a new string at the end using append().**
**Code**

```
items = ['Sumit', 'Mane']
items.append('Python')
print(items)
```

**Output**

```
['Sumit', 'Mane', 'Python']
```

**3. Insert an element at the 3rd position using insert().**
**Code**

```
items = [10, 20, 40, 50]
items.insert(2, 30)
print(items)
```

**Output**

```
[10, 20, 30, 40, 50]
```

**4. Create a list and remove a specific element using remove().**
**Code**

```
items = [10, 20, 30, 40]
items.remove(30)
print(items)
```

**Output**

```
[10, 20, 40]
```

**5. Remove the last element of a list using pop().**
**Code**

```
items = [10, 20, 30]
items.pop()
print(items)
```

**Output**

```
[10, 20]
```

**6. Remove an element from a specific index using pop(index).**
**Code**

```
items = [10, 20, 30]
items.pop(1)
print(items)
```

**Output**

```
[10, 30]
```

**7. Create a list and clear all elements using clear().**
**Code**

```
items = [1, 2, 3]
items.clear()
print(items)
```

**Output**

```
[]
```

**8. Copy one list into another using copy().**
**Code**

```
a = [1, 2, 3]
b = a.copy()
print(b)
```

**Output**

```
[1, 2, 3]
```

**9. Count how many times a specific element appears using count().**
**Code**

```
items = [1, 2, 2, 3, 2]
print(items.count(2))
```

**Output**

```
3
```

**10. Sort a list of numbers in ascending order using sort().**
**Code**

```
items = [4, 1, 3, 2]
items.sort()
print(items)
```

**Output**

```
[1, 2, 3, 4]
```

**11. Sort a list of numbers in descending order using sort(reverse=True).**
**Code**

```
items = [4, 1, 3, 2]
items.sort(reverse=True)
print(items)
```

**Output**

```
[4, 3, 2, 1]
```

**12. Reverse the elements of a list using reverse().**
**Code**

```
items = [1, 2, 3]
items.reverse()
print(items)
```

**Output**

```
[3, 2, 1]
```

**13. Find the index of a specific element using index().**
**Code**

```
items = ['a', 'b', 'c']
print(items.index('b'))
```

**Output**

```
1
```

**14. Extend a list by adding elements of another list using extend().**
**Code**

```
a = [1, 2]
a.extend([3, 4])
print(a)
```

**Output**

```
[1, 2, 3, 4]
```

**15. Create a list and check membership using in keyword.**
**Code**

```
items = ['Sumit', 'Mane']
print('Sumit' in items)
```

**Output**

```
True
```

**16. Create a list of numbers and find max and min values.**
**Code**

```
nums = [12, 5, 30, 8]
print('Max:', max(nums))
print('Min:', min(nums))
```

**Output**

```
Max: 30
Min: 5
```

**17. Create a list of mixed data types and display them.**
**Code**

```
items = ['Sumit', 20, 85.5, True]
print(items)
```

**Output**

```
['Sumit', 20, 85.5, True]
```

**18. Take user input to create a list and display it.**
**Code**

```
items = input('Enter items separated by spaces: ').split()
print(items)
```

**Output**

```
Enter items separated by spaces: Sumit Python BCA
['Sumit', 'Python', 'BCA']
```

**19. Create a list of 10 numbers and remove duplicates.**
**Code**

```
nums = [1, 2, 2, 3, 4, 4, 5, 5, 6, 6]
print(list(dict.fromkeys(nums)))
```

**Output**

```
[1, 2, 3, 4, 5, 6]
```

**20. Create a list and slice the first 5 elements.**
**Code**

```
items = [1, 2, 3, 4, 5, 6, 7]
print(items[:5])
```

**Output**

```
[1, 2, 3, 4, 5]
```

**21. Create a list and slice the last 3 elements.**
**Code**

```
items = [1, 2, 3, 4, 5]
print(items[-3:])
```

**Output**

```
[3, 4, 5]
```

**22. Create a list and replace an element at a given index.**
**Code**

```
items = [10, 20, 30]
items[1] = 99
print(items)
```

**Output**

```
[10, 99, 30]
```

**23. Create a list and find the sum of elements.**
**Code**

```
nums = [10, 20, 30]
print(sum(nums))
```

**Output**

```
60
```

**24. Create a list and find the average of elements.**
**Code**

```
nums = [10, 20, 30]
print(sum(nums) / len(nums))
```

**Output**

```
20.0
```

**25. Create a list and check if it is empty.**
**Code**

```
items = []
print('Empty' if not items else 'Not empty')
```

**Output**

```
Empty
```

**26. Create two lists and merge them into one.**
**Code**

```
a = [1, 2]; b = [3, 4]
print(a + b)
```

**Output**

```
[1, 2, 3, 4]
```

**27. Create a list and convert it into a tuple.**
**Code**

```
items = [1, 2, 3]
print(tuple(items))
```

**Output**

```
(1, 2, 3)
```

**28. Create a tuple and convert it into a list.**
**Code**

```
items = (1, 2, 3)
print(list(items))
```

**Output**

```
[1, 2, 3]
```

**29. Create a list and delete an element using del keyword.**
**Code**

```
items = [1, 2, 3]
del items[1]
print(items)
```

**Output**

```
[1, 3]
```

**30. Create a list of strings and sort alphabetically.**
**Code**

```
items = ['Mango', 'Apple', 'Banana']
items.sort()
print(items)
```

**Output**

```
['Apple', 'Banana', 'Mango']
```

**31. Create a list of numbers and sort without using sort() (use sorted()).**
**Code**

```
nums = [4, 1, 3, 2]
print(sorted(nums))
```

**Output**

```
[1, 2, 3, 4]
```

**32. Create a list and reverse it without using reverse() (use slicing).**
**Code**

```
items = [1, 2, 3]
print(items[::-1])
```

**Output**

```
[3, 2, 1]
```

**33. Create a list and find the second largest number.**
**Code**

```
nums = [10, 20, 40, 30]
print(sorted(set(nums))[-2])
```

**Output**

```
30
```

**34. Create a list and find common elements between two lists.**
**Code**

```
a = [1, 2, 3]; b = [2, 3, 4]
print(list(set(a) & set(b)))
```

**Output**

```
[2, 3]
```

**35. Create a list and remove all even numbers.**
**Code**

```
nums = [1, 2, 3, 4, 5, 6]
print([n for n in nums if n % 2 != 0])
```

**Output**

```
[1, 3, 5]
```

**36. Create a list and remove all odd numbers.**
**Code**

```
nums = [1, 2, 3, 4, 5, 6]
print([n for n in nums if n % 2 == 0])
```

**Output**

```
[2, 4, 6]
```

**37. Create a list and check palindrome list.**
**Code**

```
items = [1, 2, 1]
print(items == items[::-1])
```

**Output**

```
True
```

**38. Create a list and count total elements.**
**Code**

```
items = [1, 2, 2, 3, 2]
print(items.count(2))
```

**Output**

3

**39. Create a nested list and access inner elements.**
**Code**

```
items = [[1, 2], [3, 4]]
print(items[1][0])
```

**Output**

```
3
```

**40. Create a list and demonstrate all list methods.**
**Code**

```
items = [3, 1, 2]
items.append(4); items.insert(0, 0); items.remove(1)
items.sort(); items.reverse()
print(items)
print('Count of 2:', items.count(2))
print('Index of 3:', items.index(3))
```

**Output**

```
[4, 3, 2, 0]
Count of 2: 1
Index of 3: 1
```